Shirui (Ray) WANG

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An enthusiastic and responsible educator with three years of undergraduate-level math and statistics teaching experience, dedicated to fostering effective learning environments and supporting student success in the age of AI.

Education

Ph.D. in Biostatistics Jan 2020 – May 2026 (expected)
Georgia State University Atlanta, GA, US

M.S. in Mathematics Aug 2016 – June 2018
University of California, Irvine Irvine, CA, US

B.S. in Applied Mathematics Aug 2012 – June 2016
China Agricultural University Beijing, China

Work Experience

Georgia State University Atlanta, GA
Graduate Teaching Assistant (Instructor) Aug 2022 – May 2025

- Taught elementary statistics as **primary instructor** to non-statistics major undergraduates, effectively conveying statistical concepts and approaches to students from diverse backgrounds
- Earned an average of **4.5/5** on the student evaluation and **4.4/5** on RateMyProfessors.com
- Won the **Best Graduate Teaching Assistant Award** (2025) for outstanding teaching performance

Graduate Lab Assistant Jan 2020 – Jul 2022

- Assisted instructors with grading and tutoring in Probability, College Algebra and Precalculus class

American Cancer Society Atlanta, GA
Biostatistics Research Intern, Cancer Risk Prediction and Health Disparities May 2025 – Nov 2025

- Developed **imputation** frameworks based on **EM** algorithm and **sequential quadratic programming (SQP)** algorithm, integrated with **flexible parametric survival models**, to address **missing** stage-at-diagnosis data in **SEER** datasets
- Performed a **multinomial logistic regression** analysis to study the association between **Medicaid** coverage continuity and cancer stage at diagnosis, finding that continuous coverage significantly reduced the likelihood of distant cancer

SUNY Upstate University Hospital Remote
Biostatistician Intern Dec 2023 – Jan 2024

- Conducted a retrospective analysis of sepsis outcomes from a 2^2 **factorial observational study** to evaluate the impact of orderset and note utilization on key healthcare outcomes (mortality rate, length of stay, ICU utilization, etc.)
- Performed Chi-square tests, Friedman tests, **Mixed Effects Models**, and **Poisson Regression** with **risk-adjustment** offsets, providing robust evidence that orderset and note utilization significantly reduced mortality and ICU utilization.

Georgia State University Atlanta, GA
Doctoral Researcher Aug 2021 – Dec 2025

- Developed novel statistical approaches for evaluating continuous diagnostic tests (**ROC, AUC, Youden index**) with **partially missing** gold standard verification using **Empirical Likelihood** and **bootstrap** methods
- Conducted extensive **Monte Carlo simulation** studies and performed **biomarker validation** analyses for **real-world cancer and AD datasets** in **R**, achieving accurate and robust inferences when only 36% of the subjects were verified
- **Developed an R package *rocvb***, integrating newly proposed statistical methods into an accessible toolkit used by practitioners and non-technical users

China Construction Bank Shenyang, China
Data Mining Intern July 2019 – Aug 2019

- Developed predictive models for retail banking operations, training a **Random Forest Classifier** in **Python** to assess customer responsiveness to telemarketing, achieving 93% accuracy on the testing set
- Conducted risk analysis on transaction data to detect potential financial crimes, identifying 500+ gambling accounts based on transaction patterns

Technical Skills

- **Programming and Computing**: **R** (Package, ggplot2, Shiny, Markdown, Parallel Computing), **SAS**, **Python** (Numpy, Pandas), MATLAB, SQL, C, High-performance computing clusters, JMP
- **Statistics: Regression Analysis** (GLM, Lasso/Ridge regression), **Longitudinal Data** (Mixed effects model, Repeated Measures), **Survival Analysis** (Cox model, censored data), **Missing Data** (Multiple Imputation, Propensity scores, EM, Inverse Probability Weighting), **Real-World Data** (SEER, Medicaid), **Computational Methods** (MCMC, Bootstrap), **Clinical Trials** (Sample size calculation, Power analysis, Multiplicity adjustments, Experimental Design), **Diagnostic and Biomarker studies** (ROC, AUC), Bayesian Methods, Multivariate Statistics, Meta analysis, Causal Inference
- **Machine Learning & Deep Learning**: Random Forest, XGBoost, LightGBM, SVM, K-Means, PCA, Deep Neural Networks, RNNs, GANs, LSTM, Diffusion Models, Large Language Models, Scikit-Learn, PyTorch

Publications & Working Papers

- **Wang S**, Shi S, Qin G. Interval estimation for the Youden index of a continuous diagnostic test with verification biased data. *Statistical Methods in Medical Research* 2025; 34: 796-811. doi:10.1177/09622802251322989
- Shi S, **Wang S**, Qin G. Interval estimation for three-class Youden index with verification bias. *Journal of Biopharmaceutical Statistics* 2025; 1: 1-22. doi:10.1080/10543406.2025.2549361
- Shi S, **Wang S**, Qin G. Interval Estimation for the Sensitivity of a Test to the Early Diseased Stage with Verification Bias. *Revision at Pharmaceutical Statistics*
- Jia S, **Wang S**, Qin G. Methodological Approaches for the Estimation of Confidence Intervals on Partial Youden Index under Verification Bias. *Revision at Pharmaceutical Statistics*
- **Wang S**, Shi S, Qin G. Empirical likelihood inference for the area under the ROC curve with verification biased data. *Submitted*
- **Wang S**, Qin G. Empirical likelihood-based confidence intervals for sensitivity of a continuous test at a fixed level of specificity with verification bias. *Submitted*
- Jia S, **Wang S**, Qin G. Empirical Likelihood-Based Confidence Intervals for the Partial AUC with Verification Bias. *Submitted*
- **Wang S**, Islami F, Siegel RL, Jemal A, Choudhury PP. Imputation of missing cancer stage at diagnosis accounting for stage-specific survival. *Manuscript in preparation*
- Choudhury PP, Zhao J, **Wang S**, Jemal A, Yabroff R, Islami F. Are disruptions in Medicaid coverage linked to advanced stage diagnosis of screen-detectable cancers? *Manuscript in preparation*

Conferences & Presentations

- 2025 Joint Statistical Meeting (JSM), Contributed Paper Presentation, Nashville, TN, Aug 2025
- 2025 Georgia Statistics Day, Student Poster, University of Georgia, Athens, GA, Oct 2025
- The 10th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA, May 2025
- The Duke Industry Statistics Symposium (DISS), Duke University, Durham, NC, April 2025
- 3rd Annual Graduate Conference on Research, Scholarship & Creative Activity, Student Poster, Georgia State University, Atlanta, GA, Feb 2025
- The 9th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA, May 2024
- Scientific Computing Day, Student Poster & Lightning Talk Session, Georgia State University, Atlanta, GA, Nov 2023

Awards

- Wiley M. Suttles Math Award for outstanding math graduate student, Georgia State University, 2025
- Graduate Scholar Travel Award, Georgia State University, 2025
- Best Graduate Teaching Assistant Award for outstanding teaching performance, Georgia State University, 2025
- Mathematics Achievement Award, Georgia State University, 2022
- Best undergraduate theses (4/60), Department of Mathematics, China Agricultural University, 2016
- First Prize (Beijing) in China Undergraduate Mathematical Contest in Modeling , China Agricultural University, 2014
- Second Prize (Beijing) in China Undergraduate Mathematical Contest in Modeling , China Agricultural University, 2013
- Third-class Scholarship for Outstanding Students of College , China Agricultural University, 2012 & 2014
- Blood donation volunteer & China Marrow Donor Program volunteer, 2012